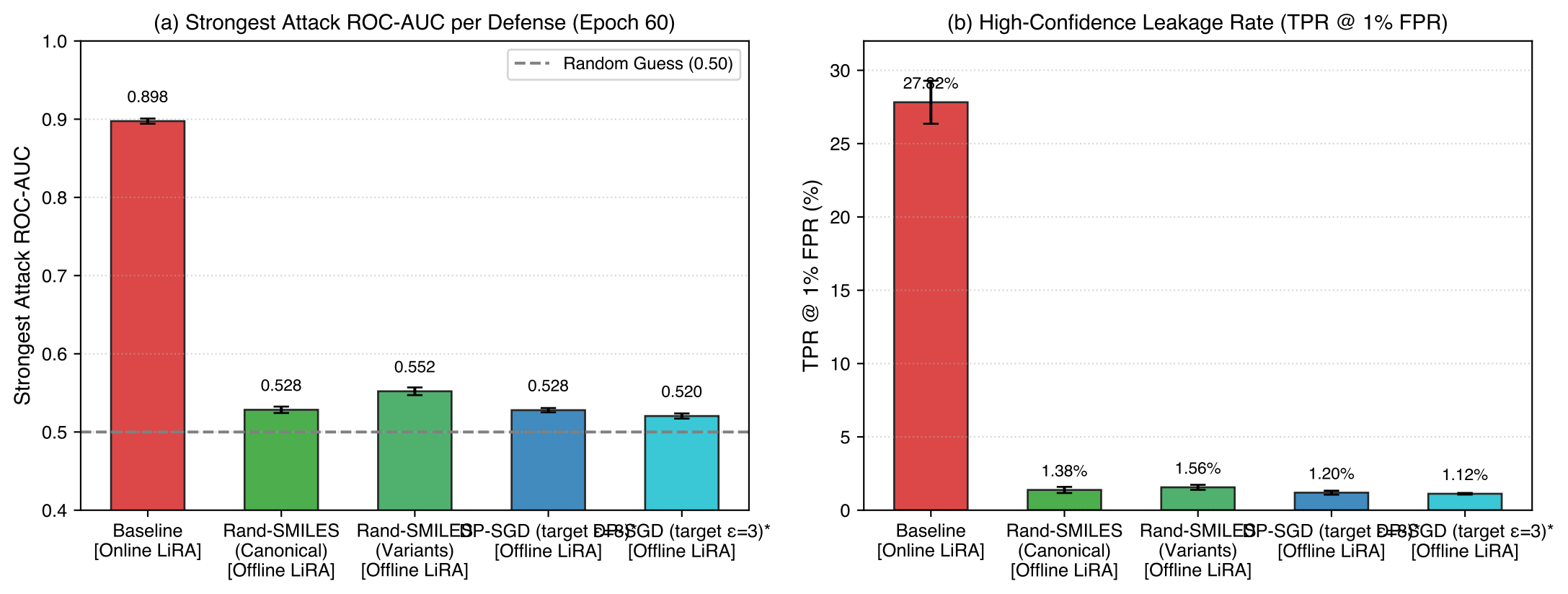
(a) Strongest Attack ROC-AUC per Defense (Epoch 60)
Strongest Attack ROC-AUC
Random Guess (0.50)
0.898
0.528
0.552
0.528
0.520
Baseline [Online LiRA]
Rand-SMILES (Canonical) [Offline LiRA]
Rand-SMILES (Variants) [Offline LiRA]
DP-SGD (target ε=8)* [Offline LiRA]
DP-SGD (target ε=3)* [Offline LiRA]
(b) High-Confidence Leakage Rate (TPR @ 1% FPR)
TPR @ 1% FPR (%)
27.82%
1.38%
1.56%
1.20%
1.12%
Baseline [Online LiRA]
Rand-SMILES (Canonical) [Offline LiRA]
Rand-SMILES (Variants) [Offline LiRA]
DP-SGD (target ε=8)* [Offline LiRA]
DP-SGD (target ε=3)* [Offline LiRA]